

Consider a membrane



The curvature free energy is an integral over the membrane surface:

$$F_c = \int dS f_c(s)$$

$\uparrow$ local curvature
 free energy.

To get f_c , expand in terms of the 2 local principle curvatures, c_1 and c_2 .

To do this, use the expression at the end of the "Taylor expansion in ω symmetric variables" notes:

$$f(\delta u, \omega) = 2k_a \delta u^2 - (k_a - k_n) \omega$$

where here, $\delta u \Rightarrow 2(H - C_0)$ and $\omega \Rightarrow K$

$\uparrow$ $\uparrow$ $\uparrow$
 "mean curvature" "spontaneous curvature" "Gaussian curvature"
 $H = (c_1 + c_2)/2$ $K = c_1 c_2$

also, $k_2 \rightarrow k_c/4$ and $-(k_0 - k_{11}) \rightarrow k_g$

"bending modulus"
aka

"bending rigidity"

"Gaussian curvature
modulus"

so, we recover the "Helfrich Hamiltonian"

$$F_c = 2k_c(H - C_0)^2 + k_g K$$

but now we need to integrate over the surface.:

$$E = 2k_c \int dS (H - C_0)^2 + k_g \int dS K$$

a constant
as long as
membrane
is closed

by the "Gauss-Bonnet"
theorem

$$F_c = 2k_c \int dS (H - C_0)^2$$